Software Engineering Lab 6: Code and Branch Coverage

Writing tests to reach full line and branch coverage with coverage and pytest

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Assigned repo: L6_L7_17_electricity_bill (Lab 6 folder)
My repo: github.com/UltraBot05/se-ps17-vaccination-scheduling, folder lab6-code-branch-coverage

1. The module

`electricity_bill.py` has three functions:

- `energy_charge(units, connection)` charges 3 per unit up to 100, 5 per unit up to 300 and 8 above that. Commercial connections pay double, then a fixed charge is added. Bad input raises ValueError.
- `late_fee(bill_amount, days_late)` gives 0 if paid on time, 50 up to 15 days late, and 100 after that, plus 5% of the bill if it is above 2000.
- `subsidy(units, is_senior, has_solar)` gives 50 to seniors using 100 units or less and 100 for solar, but the solar subsidy is removed above 300 units.

The starter file `test_electricity_bill.py` had two tests. They gave 38% line coverage and 31% branch coverage.

2. Code coverage

I added 10 tests in `CodeCoverageTest_PES2UG24CS019_electricity_bill.py`, each one aimed at lines the starter tests never reached: the two ValueError cases, the middle and top slabs, the commercial doubling, both kinds of late fee, and the three subsidy cases. With the 2 starter tests that makes 12.

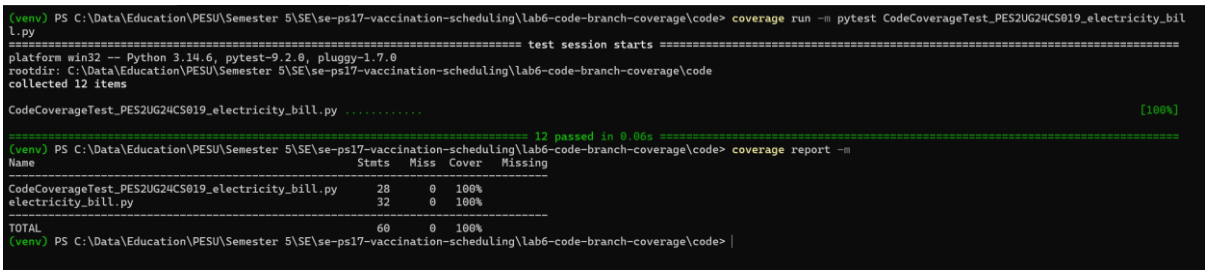


Figure 1: 12 tests pass, electricity_bill.py at 100% line coverage

Every line ran. But when I ran the same file with `--branch` it showed 98%, with one branch missing, `27->29`. In `late_fee`, once a bill is more than 15 days late, my only test used a bill of 3000. So `if bill_amount > 2000` was never false. There is no else line there, so line coverage had nothing to report as missed.

3. Branch coverage

`BranchCoverageTest_PES2UG24CS019_electricity_bill.py` keeps all 12 tests and adds 16. Only one of them was needed for the percentage, `late_fee(500, 16)`. The rest test the values where each decision flips, since that is where a wrong < or <= would hide:

Decision	Values tested	Expected
Slab edges	0, 100, 101, 300, 301 units	50, 350, 355, 1350, 1358
Commercial, top of slab 2	300 units	2750
Days late	-3, 15, 16	0, 50, 100
Bill above 2000	2000 and 2001, 20 days late	100 and 200
Senior subsidy	100 and 101 units	50 and 0
Solar subsidy	300 units, senior with solar, no subsidy	100, 150, 0

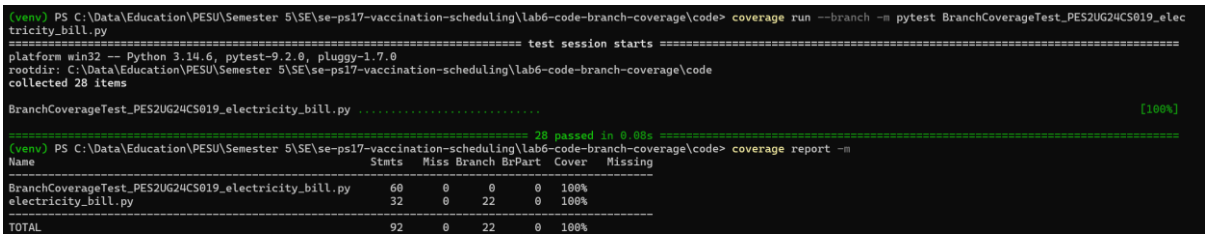


Figure 2: 28 tests pass, electricity_bill.py at 100% with all 22 branches taken

4. What I noticed

Line coverage told me my first file was done when it was not. The missed case is a real one: most late bills are small, and that was exactly the case I had not tested. Branch coverage caught it straight away.

100% branch coverage still does not mean every combination was tested. In subsidy, a senior with solar above 300 units cannot happen, because the senior rule needs 100 units or less. Path coverage would count that path, but coverage.py only measures lines and branches. Coverage also only checks that code ran, not that the answer was right. That part comes from the expected values in each assert, which I worked out by hand from the slab rules.

The TOTAL row in the reports also counts the test file itself, which is always 100%. The row that matters is `electricity_bill.py`.

5. Commits

- **Lab 6: code coverage tests, 100% line coverage** (module, starter test, code coverage file, Figure 1)
- **Lab 6: branch coverage tests, 100% branch coverage** (branch coverage file, Figure 2)
- A third commit with the README files and this document